SCAFFOLD

Knowledge-State-Driven Adaptive Scaffolding for Programming Labs

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Stream B | UN SDG 4 | FYP-1 Proposal Defense

Adaptive help needs to know what a student knows. Every student model infers that from **submission correctness** — and a student who delegates submits **correct code they do not understand**.

The workflow we are up against



4 minutes, start to finish — no struggle, no signal, no visibility

+48%	−17%	50%
practice score, unguarded AI	unaided exam score	used the bypass button

PNAS 2025, RCT, $n \approx 1{,}000$ | optional-guardrail study, $n=885$

The weakest students delegate most. **The harm is not uniform** — and uniform assistance widens the gap it is meant to close.

Where every existing system stops

System	Per-student knowledge model	Hint calibration	Leakage control	Compute / residency	Learning outcome measured
CodeHelp Koli '23	No	One-size-fits-all	Prompt guardrail	Cloud, off-prem	No
CodeAid CHI '24, n=700	No	One-size-fits-all	Pseudocode-only	Cloud, off-prem	No — usage only
ClassAid CHI '26, n=54	No	Instructor picks 1 of 4 modes	Mode-based	Cloud, off-prem	No
Codio FAST FYP '25	No — quiz difficulty	None — hands over code	None	Cloud, off-prem	No
Scaffold	**Yes — BKT / DKT**	**Automatic, per student**	**Graduated + AST**	**Local 7B, on-prem**	**Yes — transfer task**

The gap. Everyone here has assistance. **Nobody has a per-student knowledge state to calibrate it against** — and nobody keeps that state honest when the student delegates.

Codio is well built — Gemini pause-to-code, Judge0 sandbox. But every AI feature **hands the learner code**.

Codio removes friction. We calibrate it. Opposite objectives.

Three findings, three design decisions

Bastani, PNAS 2025 −17% unaided exam unguarded; −0.4% guardrailed (n.s.)	⇒	The guardrail is **load-bearing**. Ours inspects generated text **output-side**, not a line in a prompt.
DKT / Code-DKT LSTMs over (skill, correct)	⇒	Everything they observe is **correctness**. That input vector is where we intervene.
Gaming & learner modeling, 2025 up to −15.2% **AUC**, parametric worst	⇒	BKT is our **baseline to beat**, not our deployment model. Label corruption becomes an **evaluated condition**.

Every student model in this field observes only **whether the answer was right**.

Four hard constraints

C1 Adversarially corrupted estimation
Correct code is no longer evidence, and our own intervention changes the generating distribution.
No stationarity ⇒ no convergence guarantee.

C2 Real-time ordered ingestion
60 concurrent streams, $p_{95} < 200$ **ms**, zero loss. Out-of-order events reconciled per student — **a race on the mastery update**.

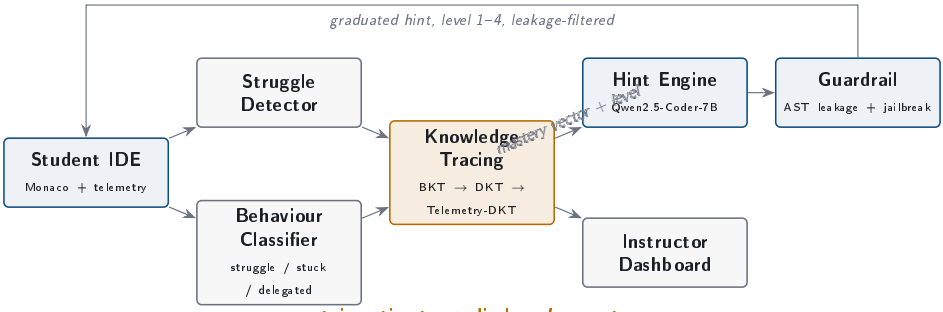
C3 Analysis of *non-compiling* C++
≈**25%** of incorrect submissions never compile — precisely the struggling students.

C4 Bounded inference budget
≤24 GB VRAM, INT4, $p_{95} < 8$ **s** at 10 concurrent, **on-premises**.

Seoul Accord CCP characteristics 1–9

Delete the language model entirely and all four problems are still there.

The closed loop



1 Closed loop — no human in the real-time path. ClassAid needs an instructor to switch modes; we compute the level.

2 Telemetry-DKT validated for *mastery validity*, not AUC — **primary claim**: does behavioural signal keep the estimate honest when the code is not the student's?

3 Graduated AST leakage measure — prior work treats leakage as binary, and all of it is mathematics.

What already runs

PROVEN TODAY

- **KT pipeline runs** — pyBKT on CSEDM / CodeWorkout, 410 students.
- **Risk experiment** — injection at 25 % / 50 %, inflation measured across all three models.
- **Telemetry live** — Monaco → WebSocket → FastAPI; vLLM + INT4 verified inside 24 GB.

POC EVIDENCE

Mastery-inflation chart
BKT / DKT / Telemetry-DKT
at 0 %, 25 %, 50 % injection

+ pyBKT AUC output
+ live telemetry capture (30 s)

`poc_screenshot.png`

NOT BUILT YET — disclosure policy · guardrail · dashboard · C++ KC taxonomy · study

If the inflation had come out flat, **we would be telling you our premise is wrong**.

How we will know

TARGETS

Primary outcome	Transfer task, *unaided*
Mastery inflation	**Telemetry-DKT $<$ DKT $<$ BKT**
DKT accuracy	AUC $\geq$ 0.70, *leave-students-out*
Capture / hint latency	$p_{95} < 200$ ms $\mid$ $p_{95} < 8$ s
Leakage	$< 5\%$ at AST sim. $\geq 60\%$
Study	$n \geq 30$, α=0.05, β=0.80

IF IT FAILS

- **Injection experiment flat** $\rightarrow$ premise disproved. We report it; the closed-loop contribution stands.
- **Qwen quality too low** $\rightarrow$ commercial API for the study only, trade-off documented, not hidden.
- **KC taxonomy slips** $\rightarrow$ transfer learning from correctness labels.
- **DPO fails** $\rightarrow$ secondary track; a reportable negative result.

Deliberately **not** time-to-solution — that metric is maximised by a system that just gives the answer.

Who owns what

Muhammad Zuhair

23I-0001 | ML

Knowledge tracing & **mastery validity**

BKT baseline · DKT comparison · injection experiment · Telemetry-DKT · DPO

Muhammad Jawad Ali

23I-3024 | Systems

Real-time capture, **latency & concurrency**

Telemetry pipeline · struggle detector · integration · instructor dashboard

Abdul Rauf

23I-0591 | Full-stack & Research

Disclosure policy, **leakage & evaluation**

Hint generation · C++ KC taxonomy · ethics · guardrail · classroom study



The two items most likely to slip — **ethics approval and the KC taxonomy** — are scheduled **first**, in month one.

DEI, ethics, legal

Equity

The weakest students delegate most. A model accurate only for strong students **widens the gap it exists to close**.

KT performance reported disaggregated by prior programming exposure — **disparity published, not omitted**.

Ethics

Consent before any collection. Ethics application **Month 1**. University-controlled storage. Grade independence in writing.

Local 7B model — **student code never leaves campus**.

Legal / IP

Qwen2.5-Coder-7B, vLLM **Apache 2.0**. pyBKT, PyTorch-DKT, Monaco **MIT**.

CSEDM under its research data-use agreement. Commercial APIs: development only, never deployed.

Scaffold never accuses anyone. It surfaces signals to a human. A false suspicion is real harm to a student — a design constraint, not an edge case.

Thank you

Questions

Adaptive help needs a student model. Every student model is computed from **submission correctness**. **Delegation breaks that measurement** — and repairing it is our contribution.

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